

Manual Camera Exposure Basics

1. Core Premise

Moving away from Auto mode grants complete creative control over how motion, depth, and ambient light are captured. Mastering manual exposure comes down to balancing the three pillars of the Exposure Triangle—Aperture, Shutter Speed, and ISO—so changes to one parameter intentionally offset the others.

2. Action Steps

1. Control Depth of Field with Aperture

Aperture (f-stops) governs the size of the lens opening. Use wide apertures (f/1.8 to f/2.8) to isolate subjects with a shallow depth of field and soft background blur (bokeh). Stop down to narrow apertures (f/8 to f/11) for sharp, edge-to-edge landscape and architectural clarity.

2. Dictate Motion with Shutter Speed

Shutter speed controls the duration the camera sensor is exposed to incoming light. Select fast shutter speeds (1/500s to 1/2000s) to freeze fast action, moving transit, or candid street subjects. Drop to slower speeds (1/30s or slower) with a stable tripod to intentionally introduce motion blur in water or night traffic.

3. Manage Sensor Sensitivity with ISO

Keep ISO at the camera's base level (typically ISO 100 or 200) in bright daylight for the cleanest image files with maximum dynamic range. Increase ISO incrementally in low-light conditions to maintain a handheld shutter speed, accepting minor luminance grain in exchange for a sharp exposure.

3. Quick Reference & Best Practices

- **Reciprocal Shutter Rule:**

When shooting handheld without optical stabilization, keep shutter speed equal to or faster than 1/focal length (e.g., at least 1/50s for a 50mm lens) to prevent camera shake.

- **Histogram Inspection:**

Use the in-camera histogram rather than the rear LCD preview to verify that highlights are not clipping against the far-right edge of the graph.